EXP 13: Case Study on Low-Code Limitations

AIM

To study the limitations of low-code development using Zoho Creator and analyze their impact on the Job Application Management System with respect to customization, performance, integration, scalability, security, and application management.

PROBLEM STATEMENT

A company wants to develop a Job Application Management System using Zoho Creator. The application should manage candidate registrations, job applications, interview schedules, document uploads, notifications, reports, and recruitment processes. Zoho Creator enables rapid application development through visual tools, predefined components, workflows, and Deluge scripting. However, as recruitment requirements become more complex, certain limitations of low-code development may affect customization, scalability, and flexibility.

SYSTEM REQUIREMENTS

- Candidate registration and login
- Job application management
- Resume and document upload
- Interview scheduling
- Email notifications
- Reports and dashboards
- Integration with external HR or recruitment systems

PROCEDURE

1. Log in to Zoho Creator and create the Job Application Management System.
2. Create forms for candidate details, job applications, interview schedules, and recruitment records using the drag-and-drop interface.
3. Configure workflows and business rules using Deluge scripting.
4. Identify limitations in advanced customization, complex business logic, third-party integrations, and UI design.
5. Test application performance with increasing numbers of candidate records and users.
6. Examine integration requirements and platform-specific dependencies.

7. Analyze the identified limitations and determine suitable solutions or alternative approaches.
8. Document the observations and conclude the suitability of low-code development for the application.

LIMITATIONS IDENTIFIED

1. Limited Customization
Zoho Creator provides predefined UI components and templates. Developing highly customized user interfaces or advanced user interactions may require custom pages or external development.

2. Complex Business Logic
Basic workflows are easy to implement. However, complex recruitment rules, multi-stage approval processes, and advanced candidate evaluation may require extensive Deluge scripting or external services.

3. Platform Dependency (Vendor Lock-in)
Applications developed in Zoho Creator depend on its platform, Deluge scripting language, and data model. Migrating the application to another platform may require redevelopment.

4. Performance and Scalability
As the number of candidates, applications, and reports increases, inefficient workflows or database queries may reduce application performance.

5. Integration Limitations
Connecting external HR systems, job portals, or third-party APIs may require API configuration, authentication, and custom integration.

6. Limited Control Over Infrastructure
Zoho Creator manages the application infrastructure, limiting direct control over server configuration, database optimization, and deployment architecture.

7. Security and Compliance
Although Zoho Creator provides security features, developers must properly configure user roles, permissions, authentication, and data-sharing settings to protect candidate information.

8. Licensing and Cost
Advanced features, additional users, and higher resource usage may increase subscription costs. Organizations should evaluate pricing before deploying large-scale applications.

PROPOSED SOLUTION

Limitation	Proposed Solution
Limited customization	Use custom pages, supported HTML/CSS, and integrations where required.
Complex business logic	Implement optimized Deluge scripts or external services.
Vendor lock-in	Document workflows and maintain exportable data backups.
Performance issues	Optimize reports, database queries, and workflows.
Integration challenges	Use supported APIs, webhooks, and connectors.
Infrastructure restrictions	Integrate external services where greater control is needed.
Security concerns	Apply role-based permissions and regularly review access settings.
Cost concerns	Evaluate licensing plans and monitor application usage.

EXPECTED OBSERVATION

S. No.	Observation	Result
1	Forms created using drag-and-drop tools	Faster application development
2	Basic workflows configured	Reduced manual coding
3	Complex logic implemented	Additional scripting may be required
4	External integrations examined	API configuration may be necessary
5	Performance requirements analyzed	Optimization becomes important as usage increases
6	Security permissions configured	Access is controlled using appropriate settings
7	Platform dependency evaluated	Migration may require redevelopment
8	Licensing requirements reviewed	Cost planning is necessary

RESULT

The limitations of low-code development in Zoho Creator were studied through the Job Application Management System case study. The analysis identified challenges related to customization, complex business logic, integration, performance, scalability, security, licensing, and platform dependency. Suitable solutions were identified to minimize their impact.

CONCLUSION

Thus, the case study on low-code limitations in Zoho Creator was completed successfully. Zoho Creator enables rapid application development, simplifies workflow automation, and reduces manual coding. However, limitations related to customization, infrastructure control, scalability, and platform dependency should be considered. For complex recruitment applications, combining Zoho Creator with custom code and external services can provide a more flexible and scalable solution.